

Computer Vision and Vitiligo: A Preemptive Diagnosis

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Problem

Vitiligo is a chronic autoimmune disorder that results in the progressive loss of pigmentation, creating visible white patches on the skin. Globally, it affects roughly 0.5–2% of the population, with profound psychological and social impacts due to its visibility and stigma.

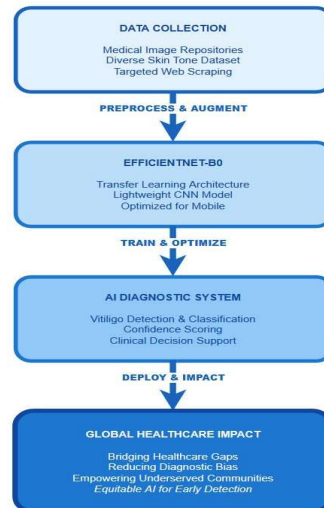


Despite its prevalence, early detection remains inconsistent, particularly in resource-limited regions where dermatology specialists are scarce. **Traditional diagnosis depends heavily on clinical expertise and visual inspection**, both of which can be subjective and prone to misclassification with conditions such as tinea versicolor or leprosy. Moreover, disparities in skin tone representation within existing medical datasets make it more difficult to detect vitiligo in darker skin types, further compounding health inequities. These gaps highlight an urgent need for accessible, accurate diagnostic tools. By leveraging machine learning and computer vision, there is potential to reduce diagnostic bias, provide more consistent results, and ultimately improve early intervention. Addressing this problem is not only a technical challenge but also a matter of equity in healthcare access and outcomes.

Purpose

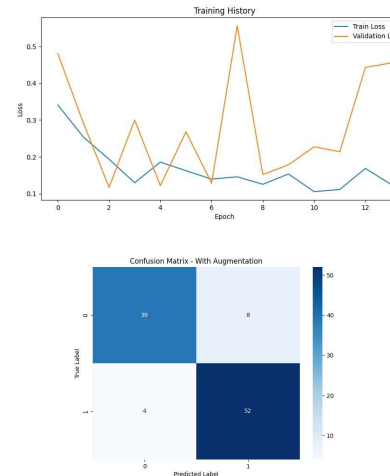
This Project develops an AI-powered diagnostic system that accurately identifies vitiligo from skin images using computer vision. The goal is to reduce reliance on in-person dermatological evaluations, making detection more scalable and accessible, particularly in regions with limited healthcare infrastructure. This system serves as an assistive tool for dermatologists rather than a replacement, demonstrating how **AI can bridge the gap between advanced medical expertise and underserved communities.**

System Architecture



Design

Integrating computer vision techniques with a robust machine learning pipeline allowed our system to classify vitiligo from digital skin images. EfficientNet-B0, a lightweight yet powerful convolutional neural network, was selected for its efficiency in balancing accuracy and computational cost. The training dataset was gathered through targeted web scraping, ensuring greater diversity across skin tones and lesion patterns. Data preprocessing included resizing, normalization, and augmentation techniques such as rotation and brightness adjustments to improve generalizability. The model was trained using transfer learning, fine-tuning pre-trained weights on ImageNet to adapt to the task-specific dataset.



Conclusion

The results demonstrate the feasibility of using computer vision and machine learning for vitiligo detection, with promising preliminary results in classification accuracy and generalizability. **These findings suggest that AI-assisted diagnosis could help address critical gaps in early detection**, especially in areas where dermatological expertise is limited. While the system is not a replacement for professional evaluation, it provides a valuable decision-support tool, reducing misdiagnosis and supporting clinicians in delivering timely care. Broader implications include improving equity in healthcare technology, as the model design actively accounts for diversity in skin tone representation—a factor often overlooked in medical AI research. Looking ahead, integrating this tool into mobile platforms could enable real-time, patient-driven screening, expanding access even further. Future work will focus on expanding the dataset, conducting clinical validation, and refining the user interface for both patients and practitioners. Ultimately, this project underscores how AI can responsibly complement medical expertise, contributing to a more accessible and equitable healthcare system.